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Covid-19 Detection based on Machine Learning (Deep Learning)

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Abstract

COVID-19 (coronavirus) is an ongoing pandemic caused by coronavirus 2 (severe acute respiratory illness) (SARS-CoV-2). The virus was initially discovered in the Hubei region of Wuhan, China, in mid-December 2019 and has since spread over the world, with more than 75.5 million confirmed cases and more than 1.67 million deaths. Given the scarcity of COVID-19 test kits in medical institutions, it's critical to design and deploys an autonomous detection system as an alternate diagnosis option for COVID-19 detection that can be employed on a commercial scale. The initial imaging tool that plays a significant role in the diagnosis of COVID-19 disease is a radiography. With radiography Images, computer vision and deep learning algorithms can aid in the detection of the COVID-19 virus. Convolutional neural networks have had a lot of success in image analysis and classification because large-scale annotated image datasets are so readily available. In this paper, we propose a deep convolutional neural network trained on open-source datasets. This research yields promising results that can be improved and commercialized.

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Chapter 1

Introduction and Background

Main points

- Introduction
- Problem Definition
- What is the current solution?
- Proposed Solution
- How will your solution solve the problem? What is new?
- Motivation
- Scope
- Methodology
- Deliverables
- Summary

1.1 Introduction:

the coronavirus (COVID-19) pandemic has afflicted billions of people Since its debut in Wuhan, China in December 2019 [1]. The virus caused a rapid spread of infection. Many studies were carried out to determine the type of virus that caused COVID-19 sickness, and it was discovered that it belonged to a large family of respiratory viruses that can cause diseases including Middle East Respiratory Syndrome (MERS-CoV) and severe acute respiratory syndrome (SARS) (SARS-CoV). Virus pneumonia can be caused by the novel SARS-CoV-2 virus. In some states, the population has had a particularly high death rate. Every day, the death toll rises around the world. As a result, a reliable, quick, and cost-effective technique for diagnosing viral pneumonia is required. Patients with COVID-19 can have no symptoms or develop pneumonia, which can lead to death in severe situations. In most cases, the virus incubates for 1 to 14 days before causing symptoms [2]. Cough, shortness of breath, fever, tiredness, and various acute respiratory distress syndromes (ARDS) have all been reported in COVID-19-positive patients [3]. The majority of people who are infected have mild to moderate viral symptoms, but they recover. Patients with severe symptoms such as acute pneumonia, on the other hand, are mainly persons over 60 years old who have diabetes, cardiovascular disease (CVD), hypertension, or cancer.

1.2 Problem Definition:

1 The global healthcare system is being overwhelmed by the exponential surge in COVID-19 patients. (As shown in Figure 3) Because of the limited number of testing kits available, it is impractical to screen every patient with a respiratory ailment using traditional Tests.

2 The traditional tests take a long time to complete and have low sensitivity.

3 Many countries lack the resources to offer these conditions for thousands of individuals to be tested. As a result, continuing to use this strategy may sluggish the pandemic's control [4].

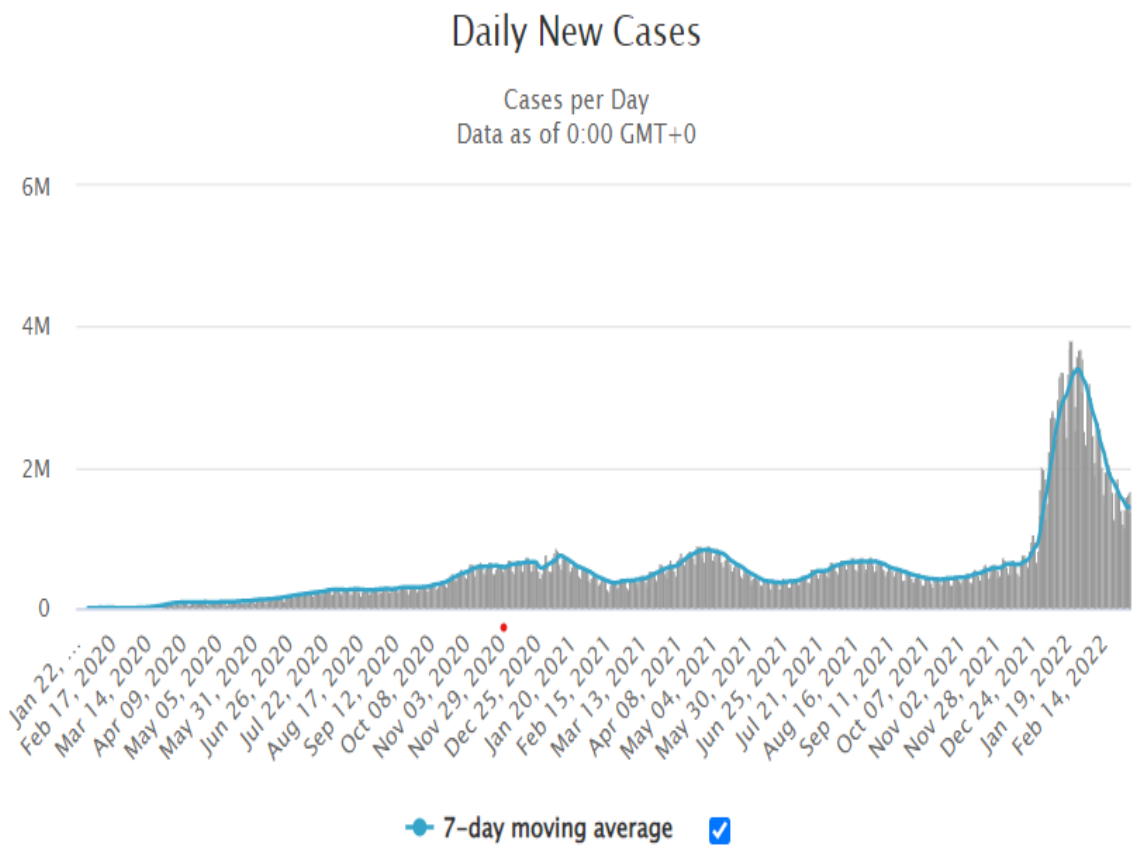

Figure 1: Covid-19 daily Cases

1.3 What is the Current Solution?

RT-PCR test. This COVID-19 test, also known as a molecular test, uses a lab technique called reverse transcription-polymerase chain reaction to detect the virus's genetic material (RT-PCR). A health care provider takes a fluid sample from the back of your nose by inserting a long nasal swab (nasopharyngeal swab) into your nostril. A shorter nasal swab (mid-turbinate swab) or a very short swab can be used to capture a sample (anterior nares swab). A lengthy swab is inserted into the back of your throat by a health care practitioner in some circumstances (oropharyngeal swab). Alternatively, you can spit into a tube to obtain a saliva sample [5].

Antigen test. The COVID-19 test looks for specific proteins in the virus. Some antigen tests can produce results in minutes using a long nasal swab to collect a fluid sample. Others may be sent to a laboratory for testing. But there's a higher risk of false-negative results, which means you could be infected with the virus but still get a negative result. A RT-PCR test may be recommended by the health care provider to confirm a negative antigen test result, depending on the situation [5].

Figure 2: Rapid Test

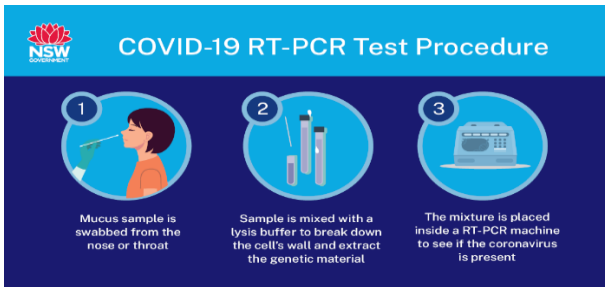

Figure 3: PCR Test

1.4 Proposed Solution:

The development of an accurate, quick, and cost-effective technique for diagnosing covid-19 is required. This will serve as the first step in implementing other outbreak-prevention measures like isolation, contact tracing, and treatment.

radiography images analysis shows promise in making the process of diagnosis faster, cheaper and more readily accessible compared to the other procedures.

Detecting suspected COVID-19 infections on a radiography could help isolate high-risk patients while waiting for test results. Most healthcare systems already have radiography machines, and because most current radiography systems are already computerized, there is no need to transport the samples. we aimed to evaluate the effectiveness of the Deep Learning Model (DLM) using radiography in screening COVID-19 cases.

1.5 How will your solution solve the problem? What is new?

When a communicable disease outbreak begins, the ideal response is for public health officials to begin testing for it early.

1. That leads to quick identification of cases, quick treatment for those people and immediate isolation to prevent spread. The model does early detection of covid-19 and this is helps to identify anyone who came into contact with infected people so they too can be quickly treated.
2. One of the advantages is that the patient will be detected and diagnosed from the home or the place where the patient is without the need to go to a doctor, and this will reduce the spread of infection.
3. Everyone can be examined and reassured without costs, except to register on our application and attach his/her chest X-ray or CT-Scan.
4. Spreading health awareness and educating the community by publishing the latest news and everything new about Covid-19, methods of prevention and protection, types of vaccines and its impact on fighting this epidemic

1.6 Motivation:

We propose a unified architecture that consists of a deep convolutional neural network (CNN) for the detection of COVID-19 from X-ray or CT Scan images, which is motivated by the urgent need to develop an artificial intelligence (AI) solution to aid in the rapid evaluation of lung diseases with COVID-19 detection, and is inspired by open-source available datasets.

Finding a model of low computational cost is important because it allows the exploitation of input images of much higher resolutions without making the processing time prohibitive. Besides, it becomes easier and cheaper to embed these models in equipment with more restrictive settings such as smartphones.

1.7 Scope:

our app is designed as a flexible, user-friendly tool for predicting Covid 19 infection in highly suspicious cases using radiographic images. This section describes the app as a mobile application's technical design, the motivation for its selection, and the Flutter framework used to develop it. We chose hybrid app development, a cross-platform application where an app developed in a common language which can run on multiple mobile platforms Android, iOS.

1.8 Methodology:

Machine learning has given computers the ability to learn on their own without having to be explicitly programmed. However, how does a machine learning system function? As a result, the machine learning Methods can be used to describe it. A machine learning project's life cycle is a cyclic process for developing an effective machine learning project. The life cycle's primary goal is to find a solution to the problem or project[6].

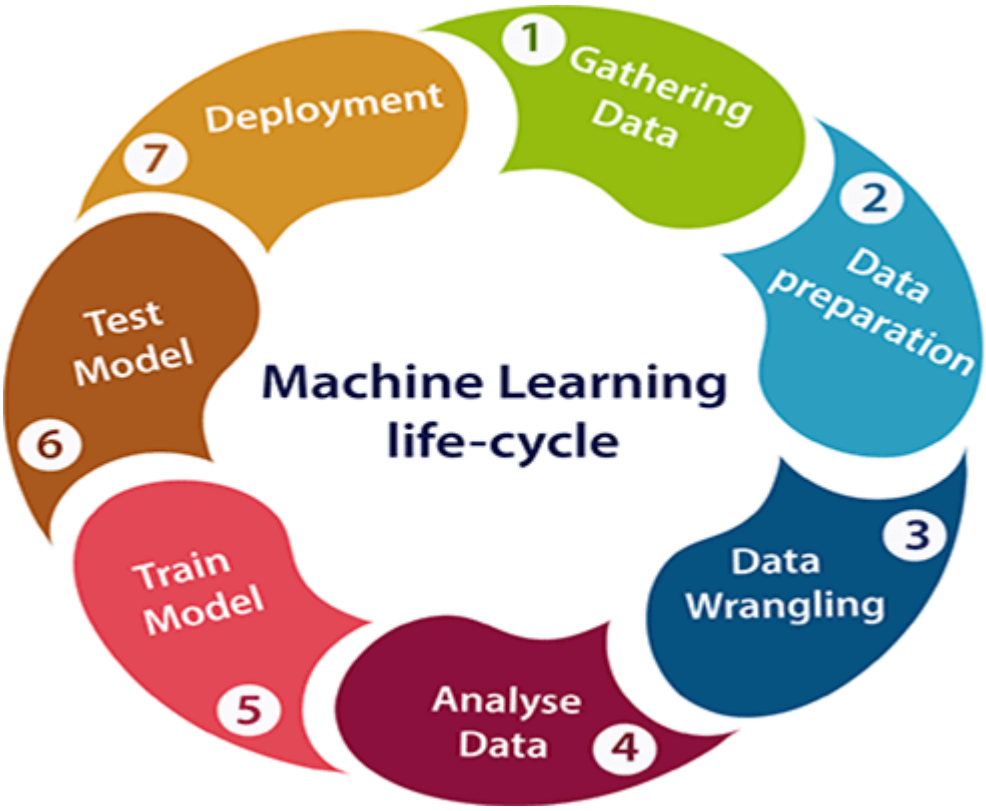

Figure 4: Machine Learning Methodologies

Understanding the problem and knowing the problem's purpose are the most important aspects of the entire process. As a result, before beginning the life cycle, we must first comprehend the problem, as a good outcome is contingent on a thorough understanding of the problem.

To solve a problem, we create a machine learning system called a "model" in the complete life cycle process, and this model is created by providing

"training." However, we need data to train a model, so the life cycle begins with data collection.

1.9 Deliverables:

1) Project Proposal
2) Covid-19 detection Machine Learning Model
3) Covid-19 detection Mobile Application
4) Project Documentation

1.10 Summary:

In this chapter, we have discussed the problem that we are going and to solve and how we are going to solve it, also we have introduced our project and mentioned the project Scope and its functionalities and what methodology we are going to use in our work, in addition to listing some of the current solution that already exist and used to solve the same problem.

Chapter 2

Related Work

Main points

- Introduction
- Related Work
- Summary

2.1 Introduction

Some of the most recent COVID-19 research calls for the use of deep learning techniques. Most researchers have been forced to use transfer learning due to the novelty of COVID-19 and the consequent lack of large data sets. By adopting three improvement procedures on the chest X-ray or CT-scan pictures we collected and using the transfer learning process of three Convolutional Neural Network (CNN) on the enhanced data set, we hope to reduce false positives and false negatives.

Transfer learning on chest X-ray or CT-scan images has been used in a number of studies to identify patients with COVID-19. We focus on those that are directly related to our proposal in this section.

2.2 Related Work:

In [7] The images were normalized in order to extract enhanced features, which were then input into deep learning image categorization algorithms CNN systems, VGG19, MobileNetV2, Inception, xception, and InceptionResNetV2 were tested to detect COVID-19 from control and pneumonia photos in a transfer scenario. The experiments were split into two parts: one with 224 COVID-19 photos, 700 bacterial pneumonia images, and 504 control images, and the other with the same normal and COVID-19 data but 714 bacterial and viral pneumonia cases. The MobileNetV2 net got the best results in the two- and three-class classifications, with 96.78 % and 94.72 % accuracy, respectively.

In [8], transfer learning was performed using VGG16, CNN and Resnet50, both of which were trained on color camera pictures from ImageNet. 10-fold cross-validation was used to determine the feasibility of using chest X-rays to diagnose COVID-19, with an overall accuracy of 89.2%.

[9] used a database of 50 controls and 50 COVID-19 cases to assess three CNN architectures (ResNet50, InceptionV3, and InceptionRes-NetV2). The accuracy of ResNet50 was 98 %.

In [10]; Successful performance in diagnosis accuracy observed in this study shows that deep CNNs could properly and effectively discriminate 21,152 normal and abnormal chest radiographs. For normal versus pneumonia categorization, the CNN model pre-trained on datasets of adult patients and fine-tuned on pediatric patients achieved an accuracy of 94.64 percent, a sensitivity of 96.5 percent, and a specificity of 92.86 percent.

In [11], four CNN networks (ResNet18, ResNet50, SqueezeNet, and DenseNet-121) were used for transfer learning. Experiments were conducted with a database that included 184 COVID-19 photos as well as 5000 no findings and pneumonia images. The sensitivity was around 98%, while the specificity was 92.9 %, according to the data.

in [12], COVID-CAPS surpasses past CNN-based models, according to the suggested model which was based on an X-ray image dataset. Despite having a far smaller number of trainable parameters than other models, COVID-CAPS achieved an accuracy of 95.7 percent, a sensitivity of 90 percent, and a specificity of 95.8%.

in [13], Individuals with COVID-19 symptoms were detected using eight different deep learning techniques from 400 chest X-ray images: VGG16, InceptionResNetV2, ResNet50, DenseNet201, VGG19, MobilenetV2, NasNetMobile, and ResNet15V2. In chest X-ray datasets, NasNetMobile outperformed all other models, with an accuracy of 93.94%.

A smart COVID-19 diagnostic model uses a barnacle mating optimization (BMO) algorithm and a cascaded recurrent neural network (CRNN) model was suggested in the research [14] employing chest X-ray images. This method finds the best possible values for CRNN hyperparameters like learning rate,

batch size, activation function, and epoch count using a set of CRNN hyperparameters. When it came time to run simulations, the BMO-CRNN model performed admirably, with a sensitivity of 97.01 percent, an accuracy of 97.31%, and an F-value of 97.53%.

Another study [15] used a CNN based on capsule networks (CapsNet) for binary and multi-class classification (COVID-19 vs. control) (COVID-19 vs. pneumonia vs. control). Experiments were carried out using a dataset comprising 1050 COVID-19 X-ray images, 1050 pneumonia images, and 1050 controls. Using a tenfold cross technique, the experiment achieved 97% accuracy for binary classification and 84% accuracy for multi-class classification.

In this study [16], a novel CNN model with end-to-end training was proposed. The study's experiments used a dataset of 180 COVID-19 and 200 normal (healthy) chest X-ray images. The study's performance was measured in terms of classification accuracy, which was 91.6%.

[17]evaluate the performance of the most recent CNN architectures employed for medical image classification in recent years, including VGG19, MobileNet v2, Xception, and Inception-ResNet-v2. Transfer learning is used by the author because it is effective in detecting numerous anomalies in tiny medical imaging datasets [18]. They studied 1,442 patient X-ray data sets, including 714 cases of bacterial and viral pneumonia, 224 cases with proven COVID-19 illness, and 504 healthy cases. The results demonstrate that among the remaining CNNs, MobileNet-v2 and VGG19 provide the most accurate categorization. While VGG19 outperforms the other techniques in terms of accuracy (achieving 98.75%), MobileNet-v2 outperforms them in terms of sensitivity and specificity (achieving 99.10 and 97.09 percent, respectively).

2.3 Summary

COVID-19 detection accuracy ranged from 89.2% to 99.1%. However, the outcome's legitimacy is questioned due to a lack of sufficient model training, especially due to the usage of such small datasets. The investigations did not take into account the compilation time of the implemented models; hence the most efficient model could not be determined with certainty. COVID-19 detection in patients necessitates not only a highly accurate but also a quick and dependable procedure.

Table 1: shows a summary of papers:

Reference	Number of Image Samples	classes	Architectures	Performance/Accuracy
[7]	**1428**	**3**	**VGG 19, MobileNetV2, Inception, Xception, Inception ResNetV2**	**Acc = 96.78%**
[8]	**204**	**2**	**VGG16 + Resnet50**	**Acc = 89.2%**
[9]	**100**	**2**	**ResNet50, InceptionV3 and Inception Res-NetV2**	**Acc = 98%**
[10]	**21.152**	**2**	**CNN**	**Acc = 94.64%**
[11]	**5184**	**2**	**ResNet18, ResNet50, SqueezeNet, DenseNet-121**	**Sensitivity = 98% Specificity = 92.9%**
[12]	**13,975**	**2**	**COVID-CAPS**	**Acc = 95.7%**

[13]	**400**	**2**	**VGG16, InceptionResNetV2, ResNet50, DenseNet201, VGG19, MobilenetV2, NasNetMobile, and ResNet15V2**	**Acc = 93.94%**
[14]	**247**	**2**	**BMO-CRNN**	**Sensitivity = 97.01%** **Acc = 97.31%** **Specificity = 97.53%**
[15]	**3150**	**3**	**CapsNet**	**Acc = 97%**
[15]	**380**	**2**	**Novel CNN Model**	**Acc = 91.6%**
[16]	**2884**	**3**	**MobilenetV2, VGG19**	**Acc=98.75%** **Sensitivity = 99.10%** **Specificity = 97.09%**
[18]	**8461 X-ray** **2481 CT**	**3**	**VGG16, DenseNet121, ResNet50, ResNet152**	**Acc=98.50%**

Chapter 3

Machine Learning Methods

Main points

- Introduction
- Dataset
- Data preprocessing and Augmentation
- Data Splitting
- Model Selection
- Performance evaluation metrics
- Summary

3.1 Introduction:

The main motivation for developing our proposed model is to automatically distinguish people with COVID-19. This study consists of three phases. In the first stage, the datasets were collected from four sources. Then we use the image Preprocessing and augmentation techniques for improving the training process efficiency. we present a detailed description of the proposed methodology designed to detect COVID-19. Finally, we apply the pre-trained CNN model to extract deep features on chest x-ray and CT-Scan images.

3.2 Dataset:

Finding an appropriate dataset is one of the criteria for image recognition and computer vision research. This study employed a dataset containing 42,330 chest X-ray (CXR) images including[20]: 7232 COVID-19, 2690 Viral Pneumonia, 12,024 Lung Opacity, and 20,384 Normal patients[21]–[28].

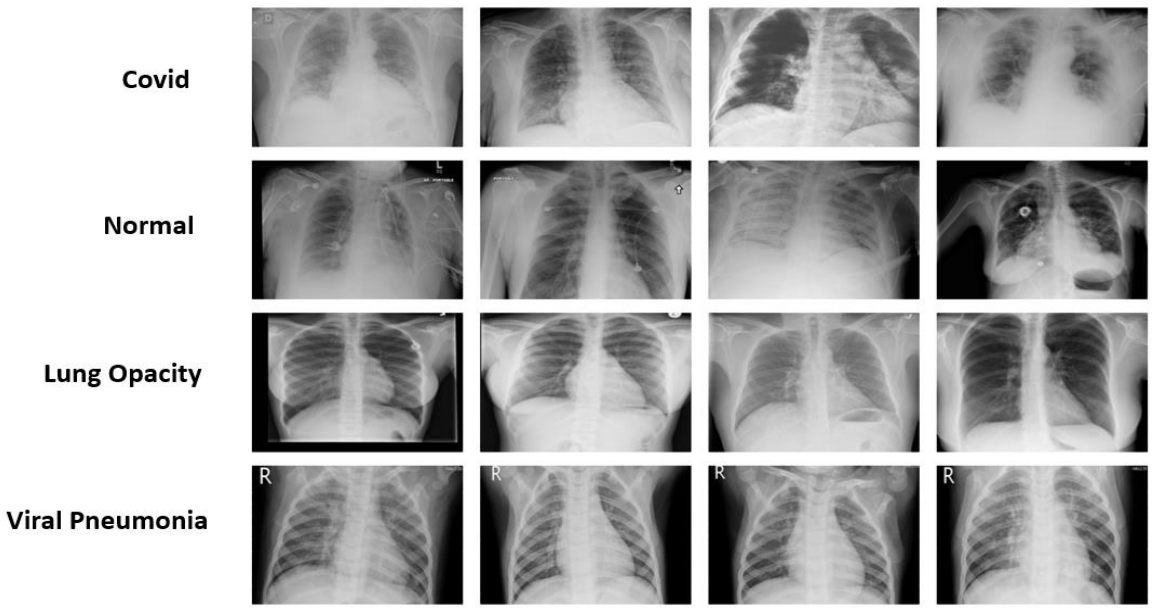

Figure 5: Example of chest X-ray Inflection images

We have also used 14,486 Chest CT scans images including[29]: 7,593 COVID-19 images from 466 patients, 6,893 normal images from 604 patients[30].

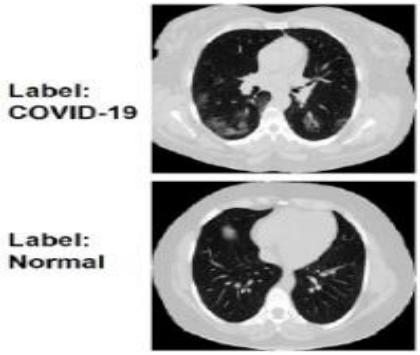

Figure 6: Example of CT-scan Inflection Images

3.3 Data Preprocessing and Augmentation:

The pre-trained model standard is used to normalize all images. **Normalization** was used to normalize the pixel values of the input images in the range of 0 to 1 for better training. One of the reasons for this is to assist with the gradient propagation problem. Normalization speeds up CNN model training and is more likely to stabilize the gradient propagation problem.

in order to avoid overfitting and increase the size of the dataset, **data augmentation** was leveraged with the following properties:

1. Randomly rotate some training images by 20 degrees.
2. Randomly shift images horizontally by 20% of the width.
3. Randomly shift images vertically by 20% of the height.
4. Randomly flip images horizontally.

3.4 Data Splitting:

The Dataset was split to 75% as training data and 25% as test data in each Class.

3.5 Model Selection:

Convolutional neural network is a class of deep neural network that is used to solve image recognition and analysis problems. The network has exceeded expectations in a number of competitions and events themed on computer vision and image analysis. Object detection and segmentation, image classification, speech recognition, and natural language processing are some of the real-world applications of the deep convolutional neural network. CNN uses multiple feature extraction at different stages to automatically learn the features represented in the dataset's image, giving it an accuracy and efficiency advantage over basic CNN.

In this section, we propose a shallow CNN structure consisting of six layers for COVID-19 classification. The bellow Figure shows the architecture of the proposed shallow CNN. Our proposed model is based on five components, namely, convolutional layer, pooling layer, dense layer, flatten layer and activation function. The details of each component of our proposed network and their respective parameters are described in next section.

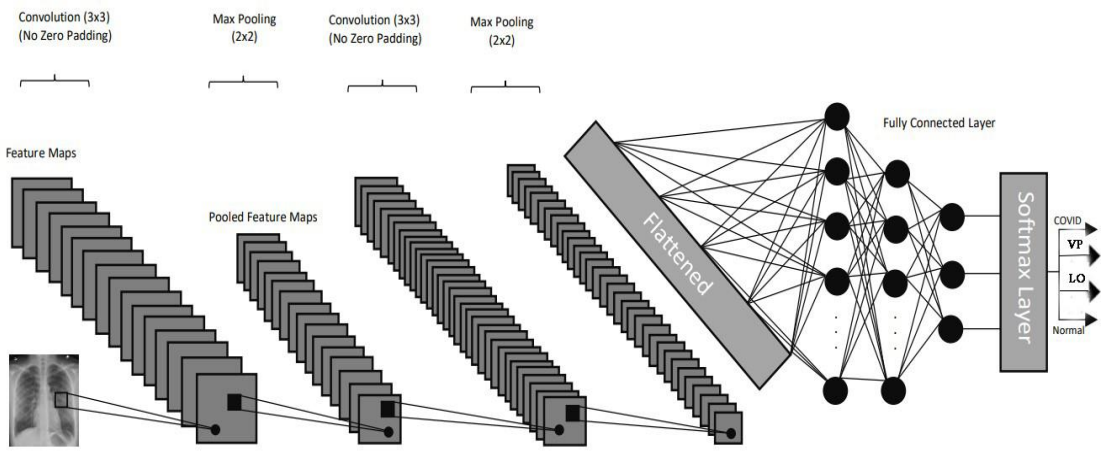

Figure 7: Architecture of proposed shallow network for X-ray

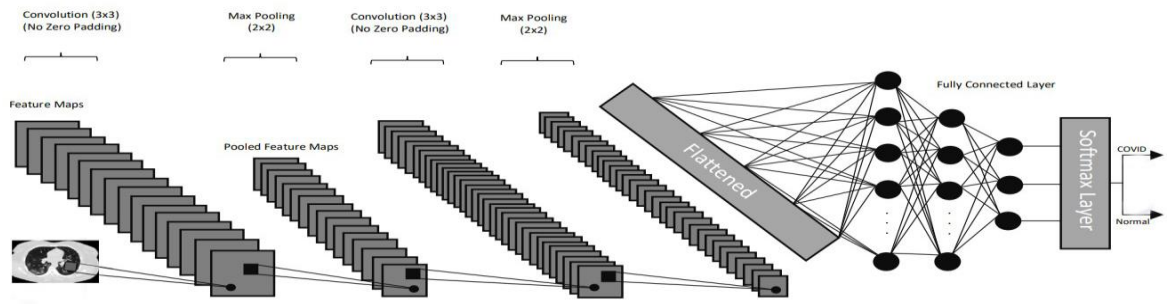

Figure 8: Architecture of proposed shallow network for CT-scan

Input layer:

The input layer basically depends on the size of the image. The size of the chest X-ray and CT Scan image has different sizes in the data set. In our network, all images must have the same size and fixed at 224×224.

Convolution layers:

The convolutional layer is the main component used in convolutional neural networks. This layer performs most of the calculations of the CNN model. It has some parameters and hyperparameters, namely, number of filters, kernel size and K. The kernel present in the filter layer is a matrix of size 3x3, which is transformed with the input features matrix.

Pooling layer:

A pooling layer is a down sampling operation, usually applied after a convolutional layer, that provides some spatial invariance. This layer can speed up the calculation and prevent overfitting. There are different types of pooling layers, including maximum pooling, sum pooling, and, average pooling. A maximum pooling layer is used in this work. It uses the maximum value of the window as output and reduces the size of the feature map. Note that we have used one max pooling layer for each convolution layer. Therefore, in our proposed model the window pooling size is 2×2.

Flatten layer:

Flattening transforms the input into a single column connecting the convolutional layer and the dense layer. In our model, we used a flattened layer after the pooling layer to flat the entire network.

Fully connected layer:

This layer determines the features that closely match a particular class by looking at the output of the previous layer. The dense layer is also called a fully connected layer. The output of a fully connected layer is provided by an activation function that predicts the class score. CNNs often use dense layers to map spatial features to image labels.

Activation functions:

The activation function adds non-linearity to the neural network and helps the neural network to learn powerful operations. The functions used are as follows:

1. We have used Rectified Linear Unit (ReLU) function with convolution layers. It provides decent calculations and avoids the vanishing gradient problem.
2. We use the softmax activation function with the dense layer to predict the class of the input X-ray and CT image.

Batch Normalization:

Batch normalization is used in CNN models to prevent and regulate overfitting, making the model much more efficient. The layer is implemented to standardize the inputs and outputs and can be used at several points in between the CNN layers. Batch normalization layer is usually placed just after the sequential model and after the convolutional and pooling layers.

Dropout:

Dropout is used to add regularization within the deep CNN network, which proves the generalization by skipping some units or connections with a certain probability at random. The dropout parameter set for the proposed model is 0.2.

3.6 Performance evaluation metrics for CT-scan:

Layer (type)	Output Shape	Param #
conv2d_16 (Conv2D)	(None, 256, 256, 32)	320
max_pooling2d_16 (MaxPooling2D)	(None, 128, 128, 32)	0
conv2d_17 (Conv2D)	(None, 128, 128, 32)	9248
max_pooling2d_17 (MaxPooling2D)	(None, 64, 64, 32)	0
conv2d_18 (Conv2D)	(None, 64, 64, 64)	18496
max_pooling2d_18 (MaxPooling2D)	(None, 32, 32, 64)	0
conv2d_19 (Conv2D)	(None, 32, 32, 64)	36928
max_pooling2d_19 (MaxPooling2D)	(None, 16, 16, 64)	0
conv2d_20 (Conv2D)	(None, 16, 16, 128)	73856
max_pooling2d_20 (MaxPooling2D)	(None, 8, 8, 128)	0
conv2d_21 (Conv2D)	(None, 8, 8, 128)	147584
max_pooling2d_21 (MaxPooling2D)	(None, 4, 4, 128)	0
conv2d_22 (Conv2D)	(None, 4, 4, 128)	147584
max_pooling2d_22 (MaxPooling2D)	(None, 2, 2, 128)	0
conv2d_23 (Conv2D)	(None, 2, 2, 256)	295168
max_pooling2d_23 (MaxPooling2D)	(None, 1, 1, 256)	0
flatten_2 (Flatten)	(None, 256)	0
dense_6 (Dense)	(None, 128)	32896
dense_7 (Dense)	(None, 64)	8256
dense_8 (Dense)	(None, 2)	130

Figure 9: Layer wise architecture of the proposed deep convolutional neural network for CT scan

The accuracy and loss of both the training and validation set were recorded for 100 consecutive epochs in **CT-scan**. The value of each category for every epoch run is shown below. It was observed that the model had an accuracy of 96.04% on the training set and an accuracy of 85.53% on the validation set as shown in

The below Figure represents the graph of Epochs versus Accuracy. Accuracies generated by the model on both the sets – training set and validation set – are marked on the graph during the 100 epochs time period

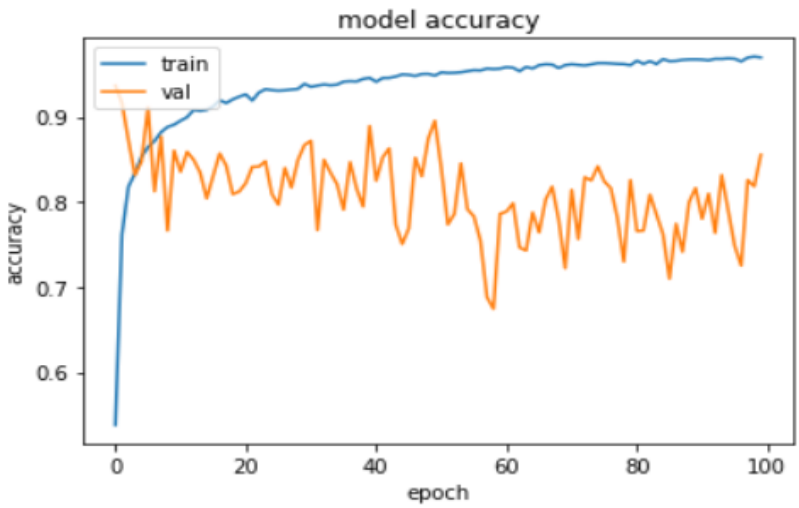

Figure 10: Accuracy obtained during 100 epochs

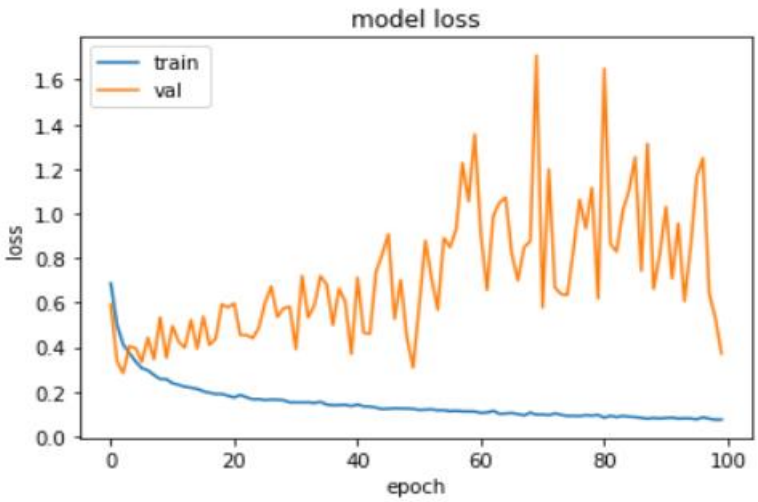

Figure 11: Graph Plotting from loss generated during 100 epochs

The ***Figure 12*** represents the graph of Epochs versus Loss. Losses generated by the model on both the sets –training set and validation set – are marked on the graph during the 100 epochs time period.

Epochs	Training Accuracy	Validation Accuracy
1	0.5381	0.9365
2	0.7618	0.9163
3	0.8179	0.8724
4	0.8335	0.8476
5	0.8530	0.8476
6	0.8644	0.9111
7	0.8716	0.8125
8	0.8819	0.8768
9	0.8881	0.7664
10	0.8954	0.8603
…. …. ….	…. …. ….	…. …. ….
91	0.9671	0.7882
92	0.9663	0.8097
93	0.9681	0.7636
94	0.9680	0.8315
95	0.9686	0.7901
96	0.9681	0.7484
97	0.9650	0.7252
98	0.9695	0.8257
99	0.9707	0.8183
100	0.9697	0.8553

Epochs	Loss	Validation Loss
1	0.6866	0.5921
2	0.5002	0.3373
3	0.4122	0.2830
4	0.3743	0.4048
5	0.3365	0.3967
6	0.3063	0.3349
7	0.2966	0.4434
8	0.2752	0.3462
9	0.2576	0.5341
10	0.2569	0.3527
…. …. ….	…. …. ….	…. …. ….
91	0.0828	1.0302
92	0.0842	0.7089
93	0.0797	0.9547
94	0.0809	0.6074
95	0.0803	0.8504
96	0.0765	1.1684
97	0.0859	1.2485
98	0.0792	0.8257
99	0.0750	0.5315
100	0.0754	0.3721

Three criteria were used for the performances of the proposed deep convolutional network as shown in the below Figure, these are Recall, F1 Score, and Precision (0 refers to Covid-19 class, and 1 refers to normal class).

	precision	recall	f1-score	support
0	0.81	0.95	0.87	1898
1	0.94	0.75	0.83	1723
accuracy			0.86	3621
macro avg	0.87	0.85	0.85	3621
weighted avg	0.87	0.86	0.85	3621

Figure 13: Performance Matrices for CT-Scan Images

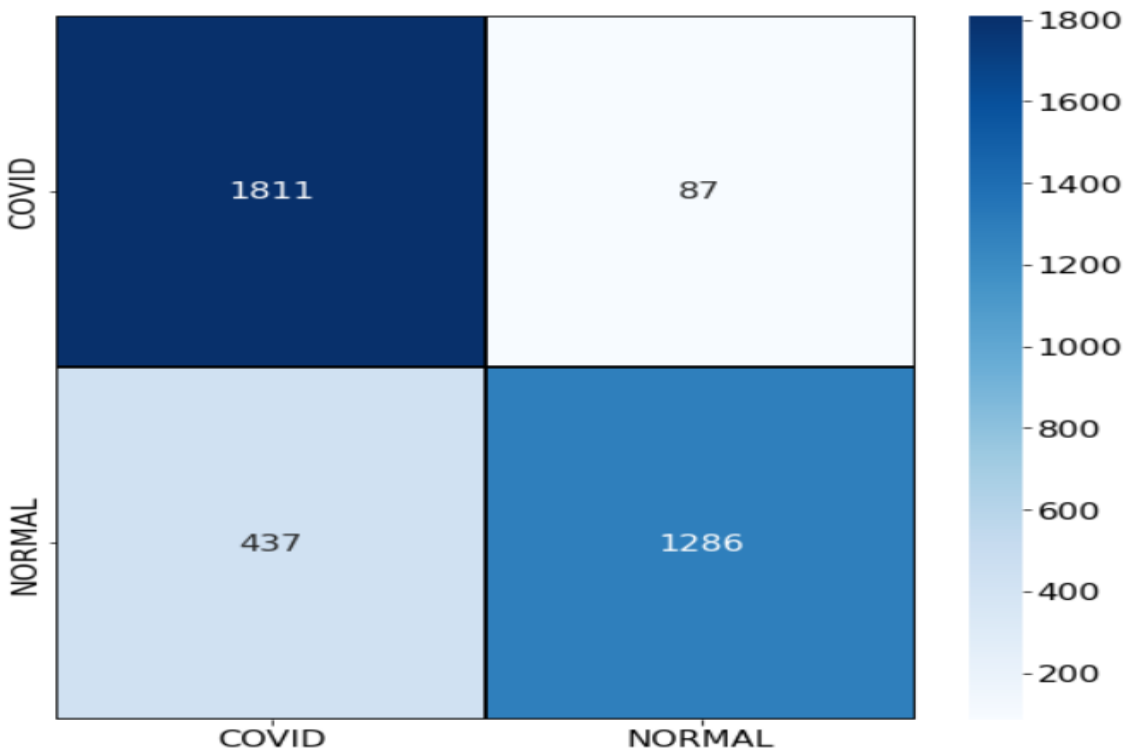

Figure 14:Confusion Matrices for CT-Scan images

3.7 Performance evaluation metrics for X-ray Images:

Layer (type)	Output Shape	Param #
conv2d_50 (Conv2D)	(None, 256, 256, 32)	320
max_pooling2d_22 (MaxPooling2D)	(None, 128, 128, 32)	0
conv2d_51 (Conv2D)	(None, 128, 128, 32)	9248
max_pooling2d_23 (MaxPooling2D)	(None, 64, 64, 32)	0
conv2d_52 (Conv2D)	(None, 64, 64, 64)	18496
max_pooling2d_24 (MaxPooling2D)	(None, 32, 32, 64)	0
conv2d_53 (Conv2D)	(None, 32, 32, 64)	36928
max_pooling2d_25 (MaxPooling2D)	(None, 16, 16, 64)	0
conv2d_54 (Conv2D)	(None, 16, 16, 128)	73856
max_pooling2d_26 (MaxPooling2D)	(None, 8, 8, 128)	0
conv2d_55 (Conv2D)	(None, 8, 8, 128)	147584
max_pooling2d_27 (MaxPooling2D)	(None, 4, 4, 128)	0
conv2d_56 (Conv2D)	(None, 4, 4, 128)	147584
max_pooling2d_28 (MaxPooling2D)	(None, 2, 2, 128)	0
conv2d_57 (Conv2D)	(None, 2, 2, 256)	295168
max_pooling2d_29 (MaxPooling2D)	(None, 1, 1, 256)	0
flatten_3 (Flatten)	(None, 256)	0
dense_7 (Dense)	(None, 128)	32896
dense_8 (Dense)	(None, 64)	8256
dense_9 (Dense)	(None, 4)	260

Figure 15: Layer wise architecture of the proposed deep convolutional neural network for X-ray

In X-ray, the accuracy and loss of both the training and validation sets were recorded for 50 epochs. For each epoch run, the value of each category is shown below. The model had a training set accuracy of 71.65 % and a validation set accuracy of 71.85.

The below Figure represents the graph of Epochs versus Accuracy. Accuracies generated by the model on both the sets – training set and validation set – are marked on the graph during the 50 epochs time period.

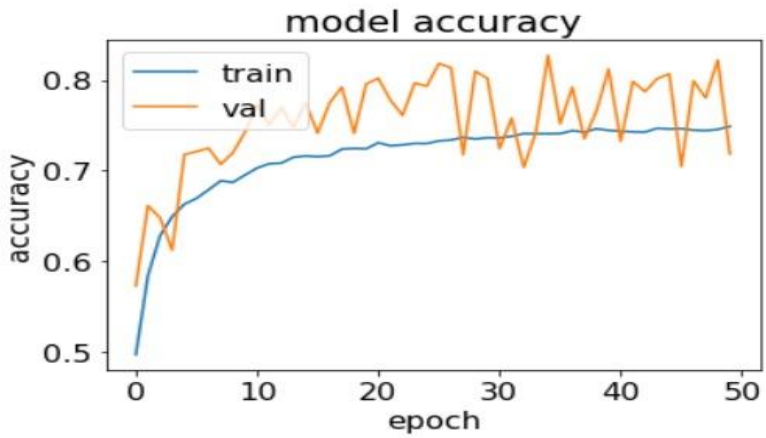

Figure 16: Accuracy obtained during 50 epochs in X-ray

The below figure represents the graph of Epochs versus Loss. Losses generated by the model on both the sets –training set and validation set – are marked on the graph during the 50 epochs time period.

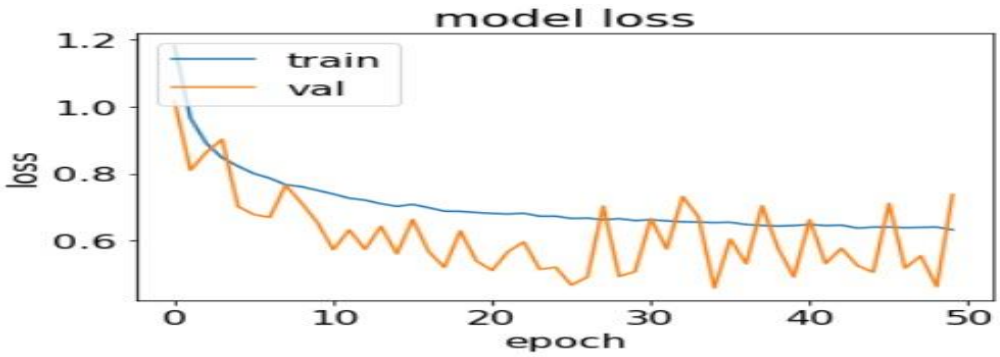

Figure 17: Graph Plotting for loss generated in X-ray Image

In X-ray, the accuracy and loss of both the training and validation sets were recorded for 50 epochs. For each epoch run, the value of each category is shown below. Table 3(a) shows that the model had a training set accuracy of 96.04 % and a validation set accuracy of 85.53%. Table 3 (b) shows the loss and validation loss generated during the experiment. The loss in the training and validation sets is 0.1011 and 0.3721, respectively.

Epochs	Training Accuracy	Validation Accuracy
1	0.4973	0.5731
2	0.5845	0.6615
3	0.6495	0.6481
4	0.6495	0.6124
5	0.6631	0.7177
6	0.6695	0.7211
7	0.6790	0.7249
8	0.6889	0.7070
9	0.6872	0.7197
10	0.6952	0.7424
…. …. ….	…. …. ….	…. …. ….
41	0.7435	0.7325
42	0.7429	0.7984
43	0.7426	0.7871
44	0.7467	0.8013
45	0.7460	0.8067
46	0.7464	0.7047
47	0.7447	0.7989
48	0.7442	0.7802
49	0.7455	0.8220
50	0.7489	0.7185

Epochs	Loss	Validation Loss
1	1.1826	1.0172
2	0.9662	0.8098
3	0.8905	0.8632
4	0.8470	0.9022
5	0.8218	0.7016
6	0.7998	0.6770
7	0.7860	0.6691
8	0.7664	0.7650
9	0.7606	0.7119
10	0.7497	0.6534
…. …. ….	…. …. ….	…. …. ….
41	0.6474	0.6617
42	0.6442	0.5304
43	0.6455	0.5762
44	0.6367	0.5247
45	0.6396	0.5046
46	0.6396	0.7105
47	0.6380	0.5168
48	0.6388	0.5535
49	0.6398	0.4615
50	0.6315	0.7373

Three criteria were used for the performances of the proposed deep convolutional network as shown in the below Figure, these are Recall, F1 Score, and Precision (0 refers to covid-19 class, 1 refers to Normal Class, 2 refers to Viral pneumonia, and 3 refers to lung opacity).

	precision	recall	f1-score	support
0	0.46	0.82	0.59	1807
1	0.85	0.74	0.79	3005
2	0.86	0.66	0.74	5095
3	0.75	0.93	0.83	672
accuracy			0.73	10579
macro avg	0.73	0.79	0.74	10579
weighted avg	0.78	0.73	0.74	10579

Figure 18: Performance Matrices for X-ray Images

3.8 Summary:

We have discussed the proposed model in this chapter to detect Covid-19 using Chest X-ray and CT-Scan datasets based on Deep learning techniques and mention all the phases which we used for transfer learning.

Chapter 4

Analysis and Design

Main points

- Introduction
- User requirements
- System requirements
- Stakeholders
- System Design
- Tools used
- Summary

4.1 Introduction:

The purpose of systems is to solve problems. The systems approach can be thought of as an organized technique of dealing with an issue. In today's fast-paced environment, the subject System Analysis and Design (SAD) is mostly deals with software development.

This chapter will explain the user and system, the system's functional and non-functional requirements. It will also identify who the stakeholders are, as well as the system design and diagrams, and the technology and tools used. To begin working on it, we created diagrams and separated the entire system into functions.

4.2 User and System Requirements:

- This system allow user to view covid-19 cases.
- This system allow user to view covid-19 Deaths.
- This system allow user to view covid-19 recovery.
- This system allow user to insert his/her X-ray or CT-scan test.
- This system allow user to Know if he/she has covid-19 or not, using his/her chest x-ray or CT-scan image.

4.3 Function requirement:

1. The user shall view covid-19 cases.
2. The user shall view covid-19 Deaths.
3. The user shall view covid-19 recovery.
4. The user shall insert his X-ray or CT-scan test.
5. The user shall know if he/she has covid-19 or not, using his/her chest x-ray or CT-scan image.

4.4 Non-Function requirement:

- Availability: The system will be available anytime, and the database can be accessed at any time of the day[31].
- Usability: the system did not need any technical knowledge to deal with, the system's interface has to be user-friendly and easy to use[31].
- Reliability: This system trains on a huge dataset which raises the accuracy of classification. So, the system should be assured that the result is precise. [32]Testing the reliability of ML systems that learn from data is difficult since failure does not always imply an error; it could just produce garbage outputs, i.e., it could provide certain outputs despite not having observed the associated ground truth during training.
- Security: The system should ensure the integrity of the user and each unsuccessful attempt by a user to access an item of data shall be recorded on an audit trail[31].
- Performance: The system should be always working properly and have a high performance by using a real-time database as a re-base because it's very ancient and it should be accessible at any time of the day so that will be available for any of the users at any time
- Manageability: When editing the code for any pages, the rest of the application stays up and running.
- Scalability: As the system grows (in data volume, traffic volume, or complexity), there should be reasonable ways of dealing with that growth[32].
- Adaptability: Other changes that typically occur in ML systems include the availability of fresh data with additional features or changes in business objectives. As a result, the system must be adaptable to rapid modifications without causing service outages[32].

- Testability: There has been work done on systematically testing the outcomes of machine learning systems. The majority of work focuses on the other hand, focuses on using machine learning to improve software testing methodologies.

4.5 Stakeholders:

- **Stakeholders:** Community, Ministry of medical**.**
- **Clients:** Health Organizations**.**
- **Users:** Hospitals or any user.

4.6 System Design:

Because logical system design is arrived at as a result of systems analysis, which is then turned into physical system design, system design is the most important phase in the creation of a system.

It is suitable to support enterprise-level client-server applications by providing solutions to scalability, security, fault tolerance, reusability, and maintainability. It helps developers to create flexible and reusable applications.

1. Usecase:

A use case diagram is a graphical description of how a system, which comprises of actors and their goals stated as use cases, would work.

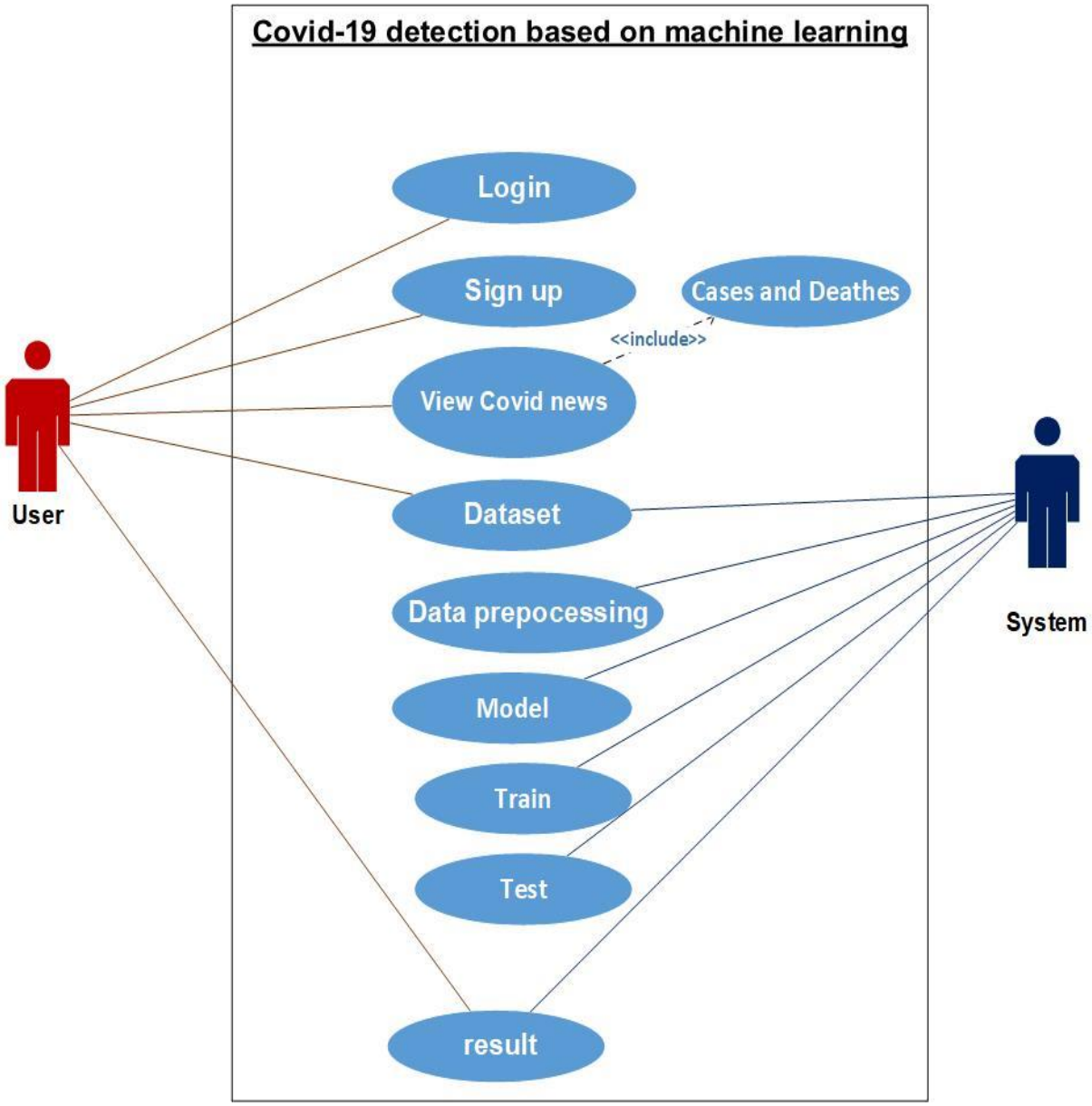

Figure 19: usecase diagram

2. Class diagram:

A class diagrams show the system's structure in terms of classes, attributes, operations, and relationships between them.

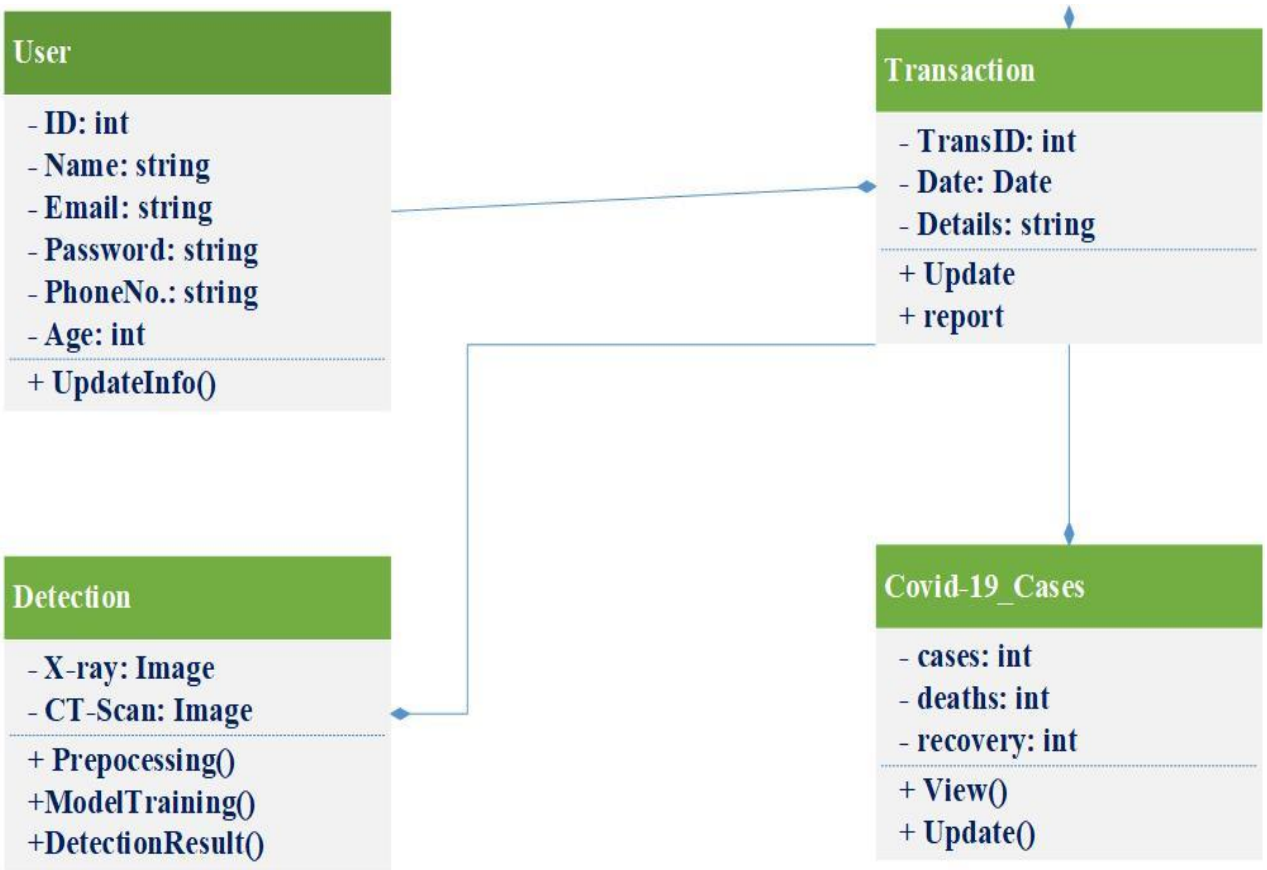

Figure 20: Class Diagram

3. Sequence Diagram:

The sequence diagram shows how each system process interacts with one another and in what order.

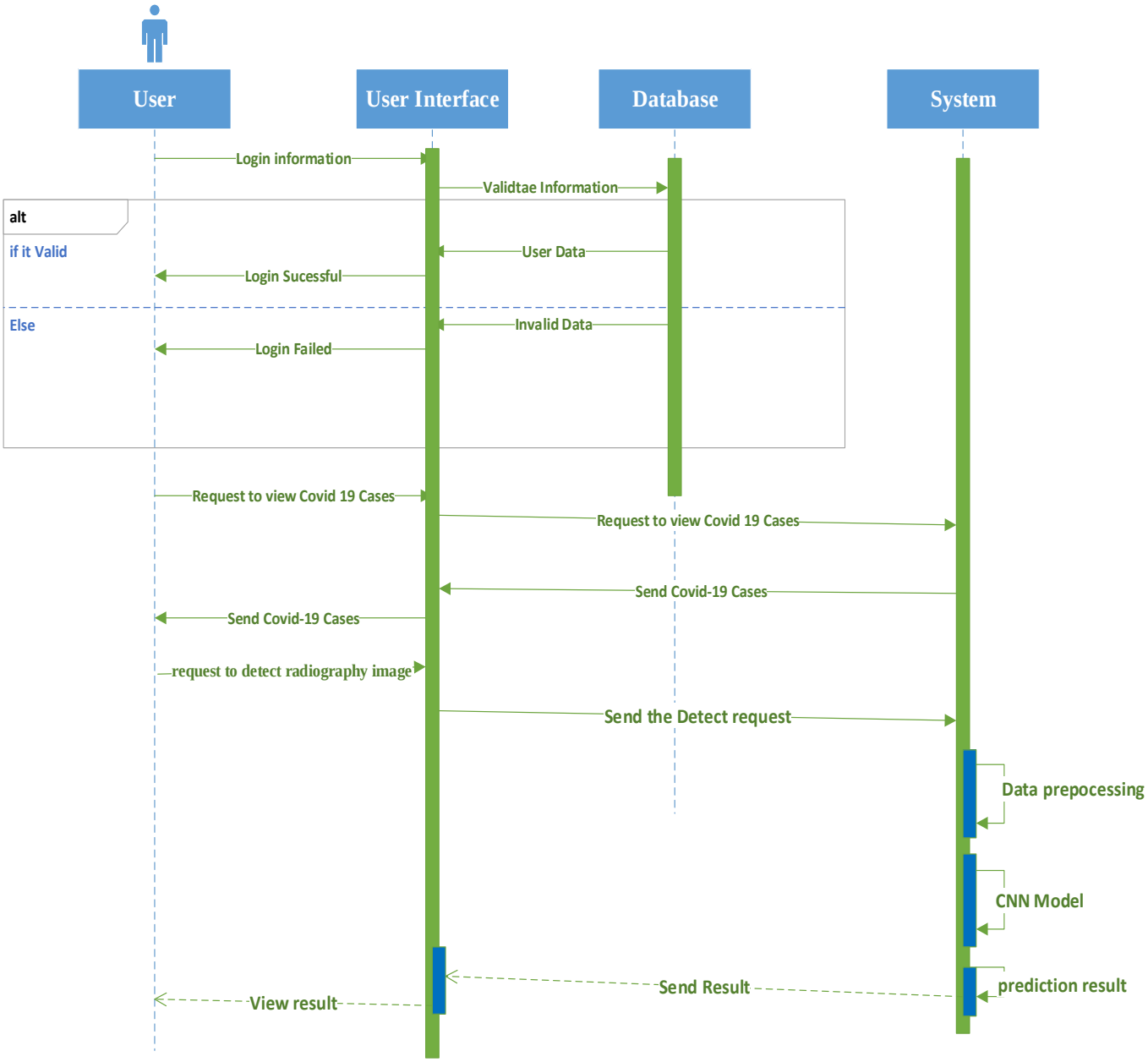

Figure 21: Sequence Diagram

4. Flowchart:

A flowchart is a picture of the separate steps of a process in sequential order.

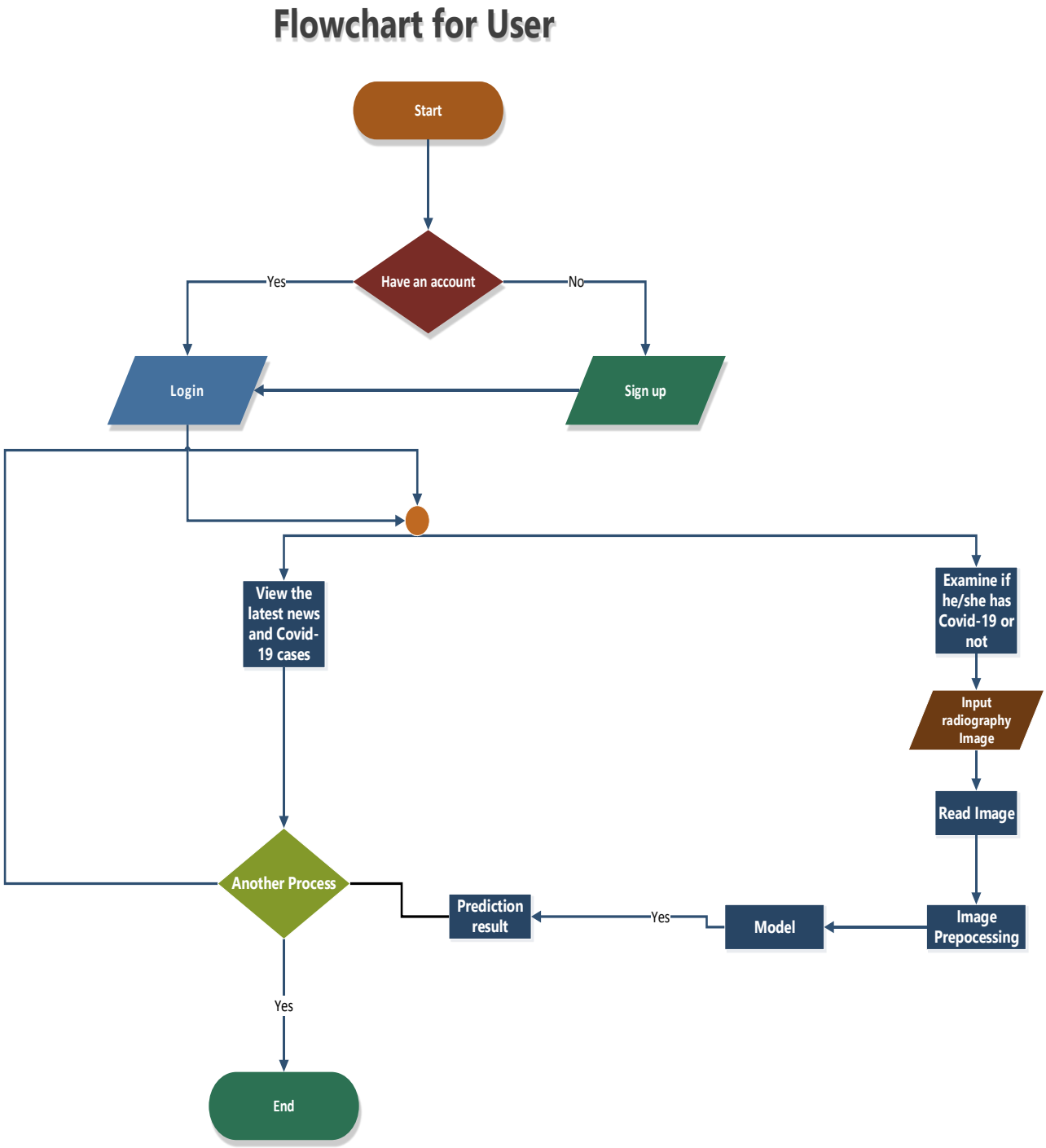

Figure 22: Flowchart

4.7 Used Technologies and tools:

1. Development Tools

- We used Android studio IDE and IntelliJ IDE for developing our Mobile application
- We used Google Colab server to code our DL model for detecting Covid-19 decease.

2. Languages

- Python 3 is used in the Google Colab server.
- Flutter framework is used in Android studio for front End.
- Dart language for back-end.

3. Components acquired from external sources

- Firebase API to build our application database.
- Tensorflow library to implement the CNN deep learning algorithm.

4. Cloud Services Provider: Google Colab.

4.8 Summary:

We reviewed our system analysis and design, the diagrams that describe it, and the tools and technologies we are utilizing. We also illustrated the functional and non-functional requirements, and the user and system requirements, in this chapter.

Chapter 5

Deliverables and Evaluation

Main points

- Introduction
- User manual
- Testing
- Evaluation
- Summary

5.1 Introduction:

We display the project deliverables and how we test our product in this phase, as well as the user manual so that he can use the project and understand its features and specifications without encountering any problems or limits. Furthermore, whether on the first demo of Future Marker or on the final output after finishing the product, we are exhibiting the user experiment and sharing the results of the experiment as well as the feedback we have received. Also Explain performance metrics that will be used to quantify the quality of a solution to the problem.

5.2 User manual:

As mobile phones are available widely, which can be a source for collecting data from the people, which includes their symptoms, travel history and behavior by which they can inform the risk that the people may face. A mobile based application is proposed which uses AI based algorithm that runs on cloud and it helps the people in order to identify the cases of COVID-19.

our app is designed as a flexible, user-friendly tool for predicting Covid 19 infection in highly suspicious cases using radiographic images. This section describes the app as a mobile application's technical design, the motivation for its selection, and the Flutter framework used to develop it. We chose hybrid app development, a cross-platform application where an app developed in a common language which can run on multiple mobile platforms Android, iOS.

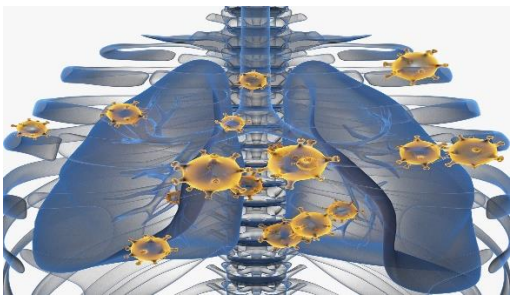

Figure 23: our application logo

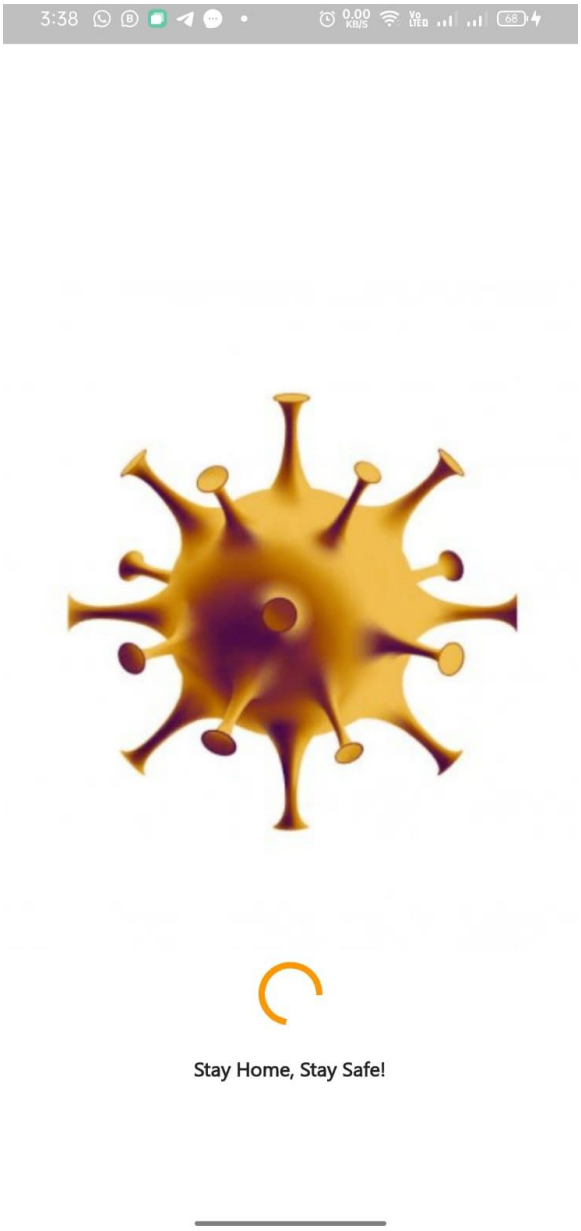

Figure 24: Splash Screen

A **Splash screen:** is a graphical control element that consists of a window with an image, a logo, and the software's current version. While an application or software is starting up, a splash screen is launching.

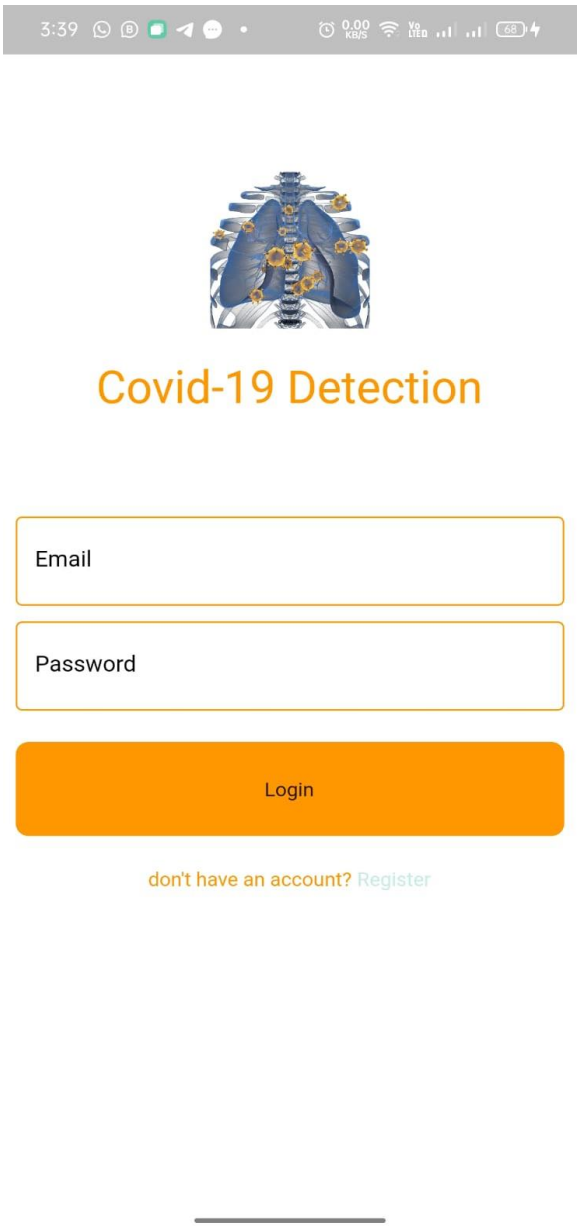

Figure 25: Login page

Login screen: user can log in with (1) email and (2) password. if he didn't have an email, navigate to the signup screen by clicking the (3) register

(6) finally, he can log in by clicking the (5) login button.

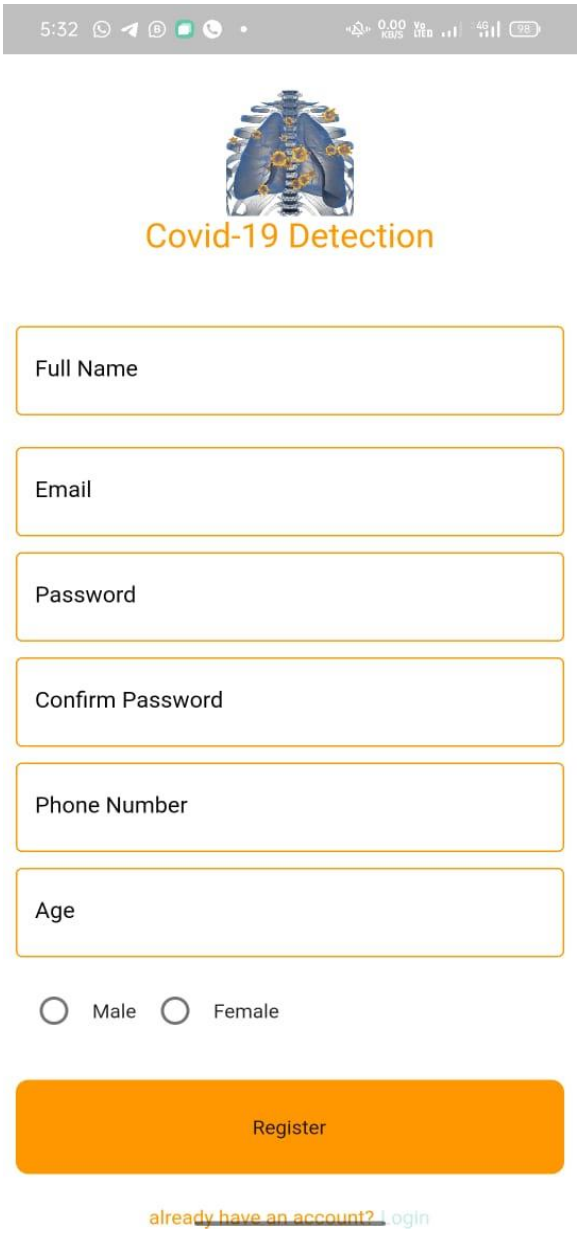

Figure 26: registration page

Registration screen: where the user can create his account by adding his (1) full name then adds his (2) email to use it to log in with the (3) password, (4) confirmation password, adding a user (5) phone number and choosing the (6) gender is the final step before submitting his registration form by click (7) register button.

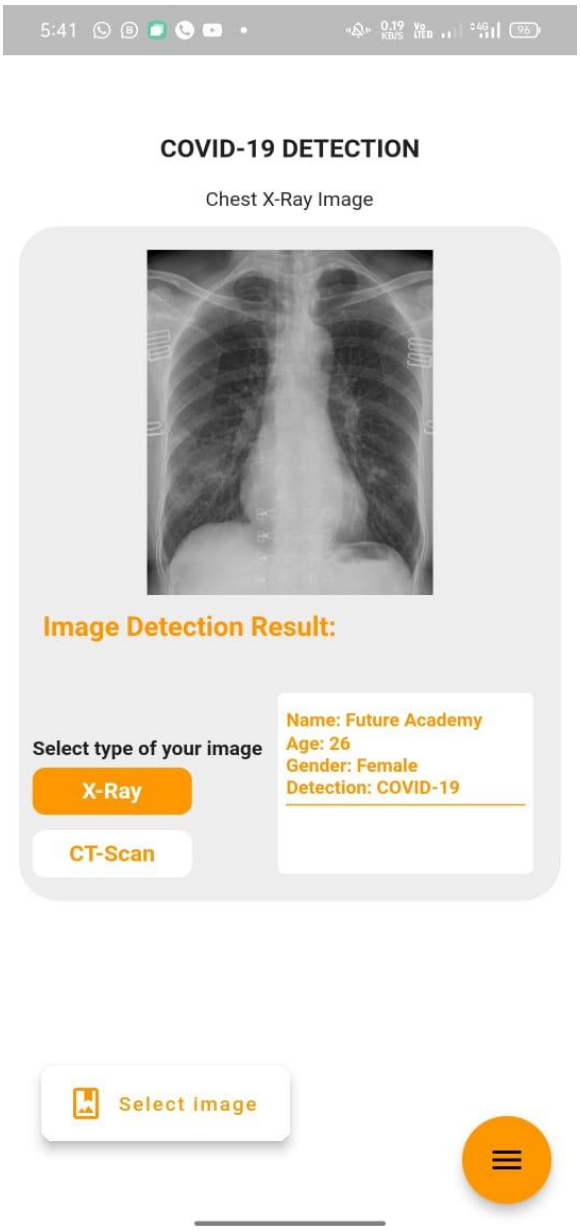

Figure 27: Home page

Detect page: where the user can choose (1) X-ray or CT-Scan button then adds his (2) radiography image, (3) the user see the scan result, see an option menu button to navigate between pages.

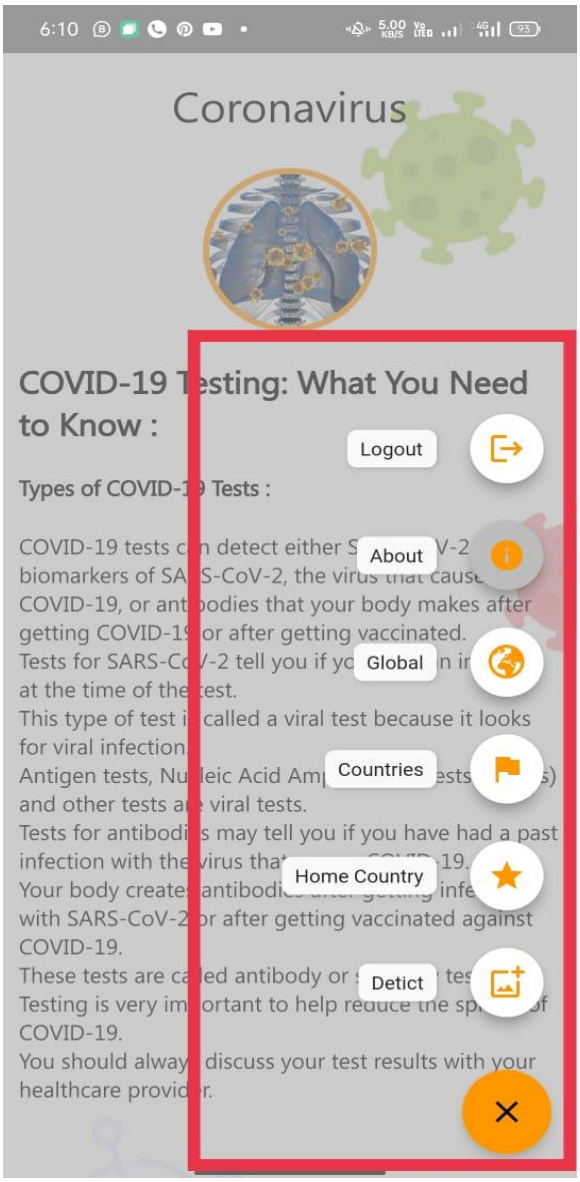

Figure 28: After pressing option menu button

Option menu button: (1) Logout button when the user pressing this button, he will logout and return to the login page. (2) About button will navigate the user to the About us page. (3) when the user presses the Global button; he will navigate to global COVID-19 page which have the total global cases, deaths and recoveries. (4) when the user presses the Countries button; he will navigate to countries page which have the total cases in Countries and for details he any choose any country to see today cases, total cases, total deaths, today deaths, Total tests, and Critical cases in countries. (5) Home country will navigate the user to his/her country to see details in his/her country. (6)Detect will let user insert CT-Scan or X-Ray images to detect diseases.

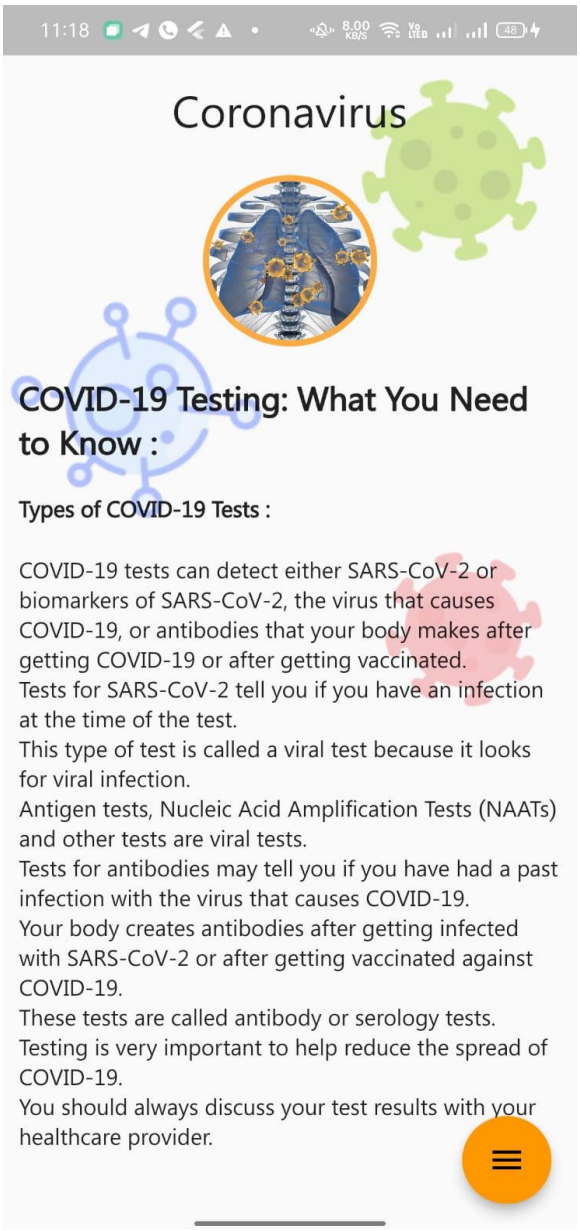

Figure 29: Page about Corona virus

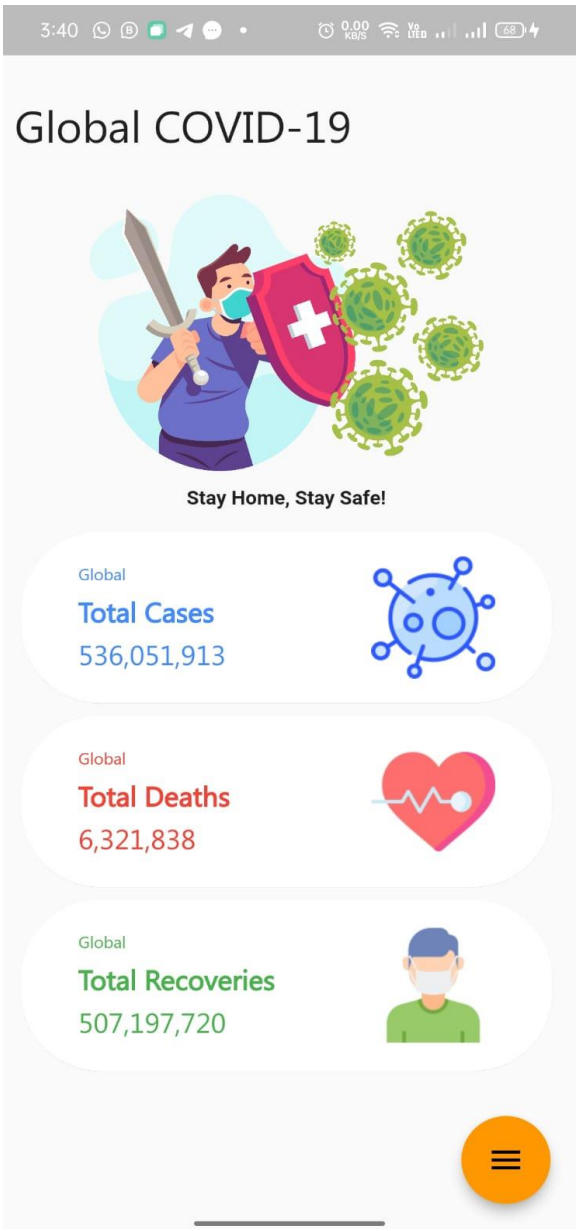

Figure 30: Global Covid-19 page

Figure 31: Home country page

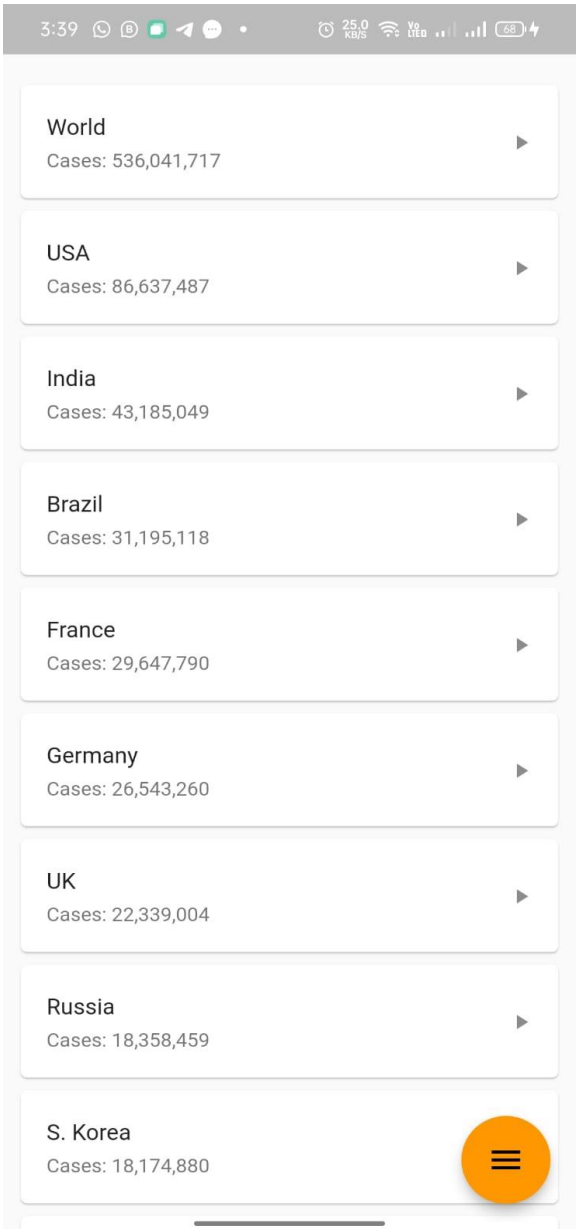

Figure 32: Countries page

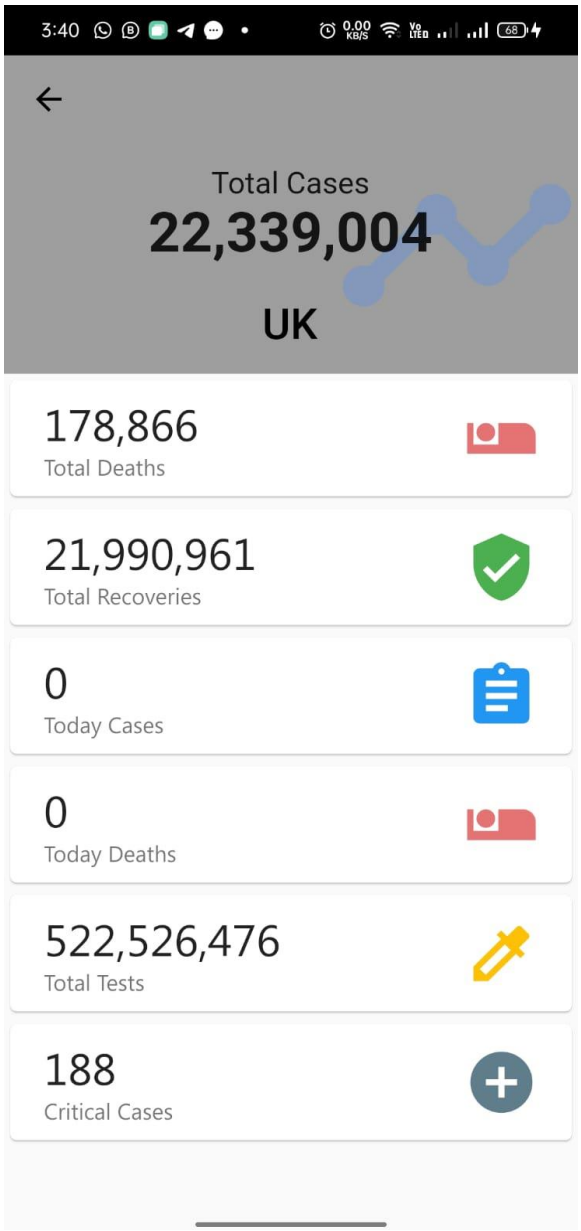

Figure 33: if he chooses any country

5.3 Testing:

No	Module	Test case Description	Result
1	Installation	Verify that application can be Installed Successfully.	Can be installed
2	Uninstallation	Verify that application can be uninstalled successfully.	Can be uninstalled
3	Splash	A splash screen is a graphical control element consisting of a window containing an image, a logo, and the current version of the software. A *splash screen* .	When user starts the application the manager splash should be Displayed without errors
4	Application Start/ Restart	1. Find the application icon and select it 2. Press a button on the device tolaunch the app 3. Observe the application launch in the timeline defined.	Application takes 23s to start
5	Log in	Verify that you can login with existing account.	You can login with your account
6	Navigation	You can navigate between the pages with right order.	You can navigate between the pages with right order without error
7	Display the Cases, Deaths and recoveries	You can see the total and the Daily number of cases and Deaths and recoveries.	This option is running successful without errors.
8	Detection	You can test if you have Covid-19 or not using X-ray and Ct-scan Image	The accuracy of Detection with X-ray is 71.65% The accuracy of Detection with CT-Scan is 96.04%

5.4 Evaluation (User Experiment):

After finishing our main functions as a result of this experiment some users reported us about their experience which involve some of the following:

1. They asked us to add CT-scan for detect covid-19.
2. They asked us to Increase the epochs number in the model training.

After receiving feedback, we started searching about how we can fix them and we successfully did it all problems have been resolved and we did some improvement which includes:

1. Adding CT-scan for detect covid-19.
2. Increasing the epochs number in the model training.

5.5 Summary

In this part we have discussed how we tested our projects and what are the results of this testing and how the received feedback from the users helped us to maintain and fix the bugs, in addition to we are showing the User Manual for Future Marker that shows the user how to use it Furthermore, we have discussed our evaluation for the user experiments.

Chapter 6

Discussion and Conclusion

Main Points

Introduction

Main Findings

Why is this Project Important?

Practical Implementation

Future Recommendation

Conclusion Summary

6.1 Introduction:

Here in this chapter, we have reached the last phase of the project after we have finished the previous phases as we have discussed in them the main problem that we are solving and how we have solved it and what methodology we have followed, and what tools we have used, besides the deliverables of the project, also we have explained the functional and non-functional requirements, besides the user requirements, furthermore, we have shown the diagrams that clarify our system, we have also presented our testing and user experiment, and finally, here we are going to discuss the main findings that we have found and reached and why our project is very important as we believe it can be a change in online learning s and what is we are planning to do regards the future improvements and recommendations for Our Project.

6.2 Main Findings:

Current AI-based COVID recognition studies in chest CT and X-ray images are primarily focused on detecting disease in affected lungs, accurately segmenting anatomic structure and lung opacities, and classifying disease severity. This is primarily accomplished through the use of machine learning and deep learning techniques. Because they contain both feature extraction and classification stages, convolutional neural networks or different models derived from convolutional neural networks are widely used. The emphasis on AI implementation in both predictive and prognostic decision support systems is reflected in various societies' recommendations. However, the use of artificial intelligence (AI) on CT scan and X-ray data for COVID-19 screening or first-line diagnostic tests is still up for debate.

6.3 Why is this project important?

Early detection of COVID-19 is important to interrupt the human-to-human transmission of COVID-19 and patient care. Currently, the isolation and quarantine of the suspicious patients is the most effective way to prevent the spread of COVID-19. Diagnostic modalities such as chest X-ray and CT are playing an important role in monitoring the progression and severity of the disease in COVID-19 positive patients. This paper presents a feature extractor-based deep learning and machine learning classifier approach for computer-aided diagnosis of COVID-19 pneumonia. Several ML algorithms were trained on the features extracted by well-established CNNs to find the best combination of features and learners.

6.4 Practical Implementations:

The proposed method is generic as it does not need handcrafted features and can be easily adapted, requiring minimal pre-processing. The provided dataset is collected across multiple sources with different shape, textures and morphological characteristics. The transfer learning strategy has successfully transferred knowledge from the source to the target domain despite the limited dataset size of the provided dataset. During the proposed approach, we observed that no over-fitting occurs to impact the classification accuracy adversely.

This Mobile application could help doctors benefit from our proposed method by providing an app tool that only requires uploading an X-ray or CT-scan image. The application then provides the physician with a simple COVID-19 Positive, or COVID-19 Negative observation.

6.5 Limitations:

The experiments used an open-access dataset of lung images, which included COVID-19, viral pneumonia, lung opacity, and healthy-person examples. To develop a fine-tuned version of the evaluated techniques, a larger dataset would be required. To the best of our knowledge, most modern medical imaging machine learning tools have this limitation. To validate the generalization of any automated diagnostic assistant tool in the future, a classification tool trained on a large dataset of hundreds or even thousands of patients (rather than a few images taken from a few patients) will be required.

6.6 Future Recommendation:

Though we achieved extraordinary results, it is not ready yet to replace the traditional testing techniques for COVID-19 detection. however, it can be extremely useful in assisting clinicians in making proper COVID-19 detection decisions. Our model will be more effective and accurate in the future. We want to test different feature extraction techniques to see which one produces the most accurate results for our model.

6.7 Conclusion Summary:

Finally, we have summarized and discussed the challenges faced and the potential research opportunities and future directions in the area. Therefore, the challenges that are identified create promising research opportunities in the field which must be addressed with effective solutions in various application areas.

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